OpenClaw scheduled task tutorial

> Before starting, please complete [01-Peripheral hardware configuration], [02-openclaw API-KEY configuration], [03-openclaw switching model], [04-OpenClaw prompt words], [05-AI voice interaction configuration], [06-Three dialogue scheme configuration tests] in the `Preparation before use` directory. If you are currently only using TUI/WhatsApp, you can skip `05` first. By default, the basic environment of this article has been prepared and only the "automatic execution of scripts at fixed time" section will be expanded.

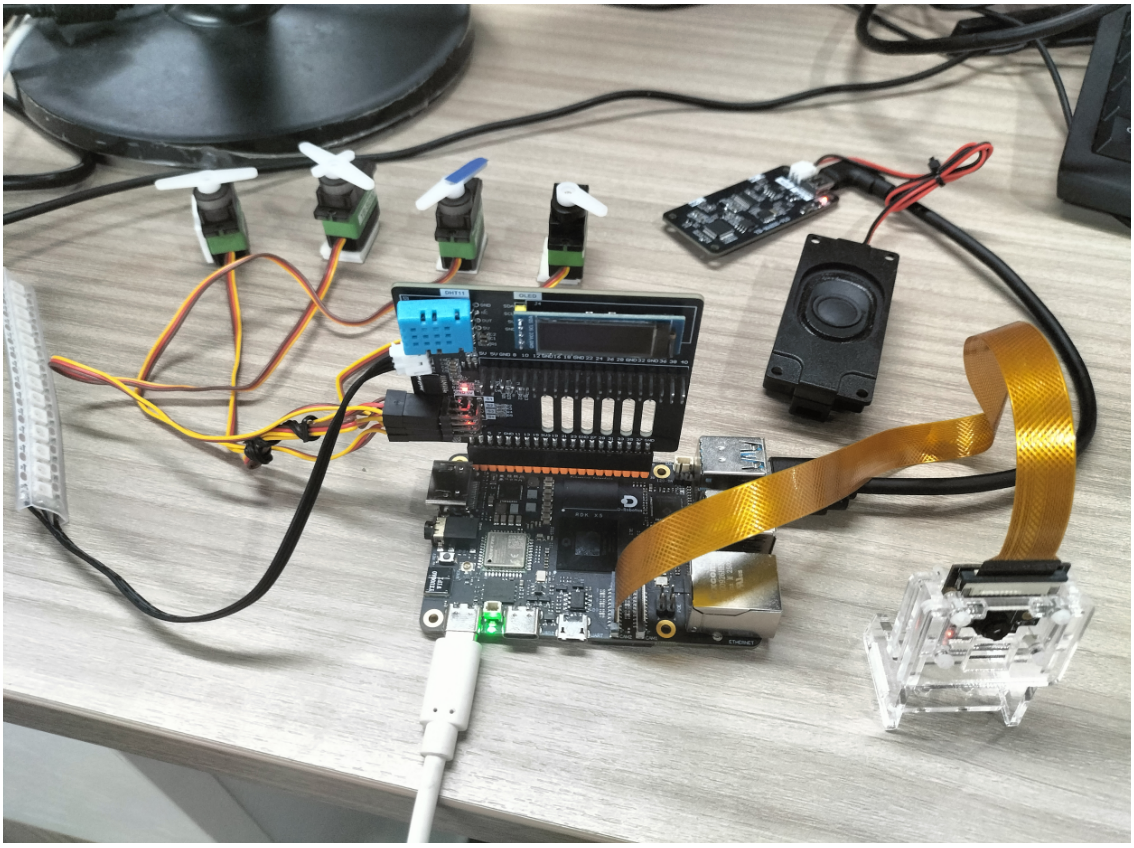
1. Tutorial Objectives

This article uses the method of "AI generated script + Linux scheduled execution" to show how OpenClaw can be further upgraded from real-time dialogue to automated peripheral applications. After completing this tutorial, readers will be able to:

- Understand the basic implementation ideas of OpenClaw scheduled tasks
- Distinguish the appropriate scenarios for `HEARTBEAT` and `cron`
- Describe automation requirements through three methods: WhatsApp, TUI, and voice code

2. Hardware preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display
- RGB light strip/RGB lamp bead module
- `[optional] CSI camera`
- `[Optional] AI voice module + speaker`



Description:

- The core of this tutorial is "scheduled execution", it is not mandatory for all peripherals to be connected at the same time
- Which type of automation tasks you want to do, just connect the corresponding hardware first

3. Test basic capabilities of scheduled tasks

Before starting the current project, please complete a basic dialogue self-test of `openclaw tui`; if you have not done so yet, please read [06-Three dialogue scheme configuration tests] first. After passing, enter the timed task special test.

It is recommended to test in the following order:

1. First enter OpenClaw TUI
2. Let OpenClaw generate a simple script
3. Run the script manually in the terminal to confirm that the function is normal
4. Then write the script into `crontab`

Terminal input:

```
openclaw tui
```

It is recommended to try the following requirements first:

- Help me write a script to display the current temperature and humidity on OLED and save it to the desktop
- Generate a script that reads temperature and humidity and prints the results each time it is run.
  -Write a script to make the RGB light turn green for 3 seconds and then turn off

After the script is generated, run it manually first, for example:

```
python3 /home/sunrise/Desktop/your_script.py
```

4. Three options for talking to OpenClaw

Solution	Suitable scenario	Advantages	Suggested positioning
WhatsApp	Remotely describe automation requirements	Suitable for clarifying "what you want to do on a regular basis" first	Used to generate scripts and confirm requirements
TUI	Create and debug scheduled tasks locally	The easiest way to jointly debug scripts and `cron`	The preferred implementation solution
Voice code	Voice description automation scene	Closer to desktop assistant experience	Suitable for demonstrating "speak a sentence to generate automation"

If the AI voice module + speaker is optional, Feishu dialogue, tui, and WhatsApp dialogue need to broadcast the content of openclaw's reply. You can run the following program on the terminal. The AI voice module dialogue does not need to be run repeatedly

- Modify whether to open broadcast parameters

```
/home/sunrise/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter is to control whether to broadcast. After modification, save it and run the following program on the terminal.
```

- Terminal input

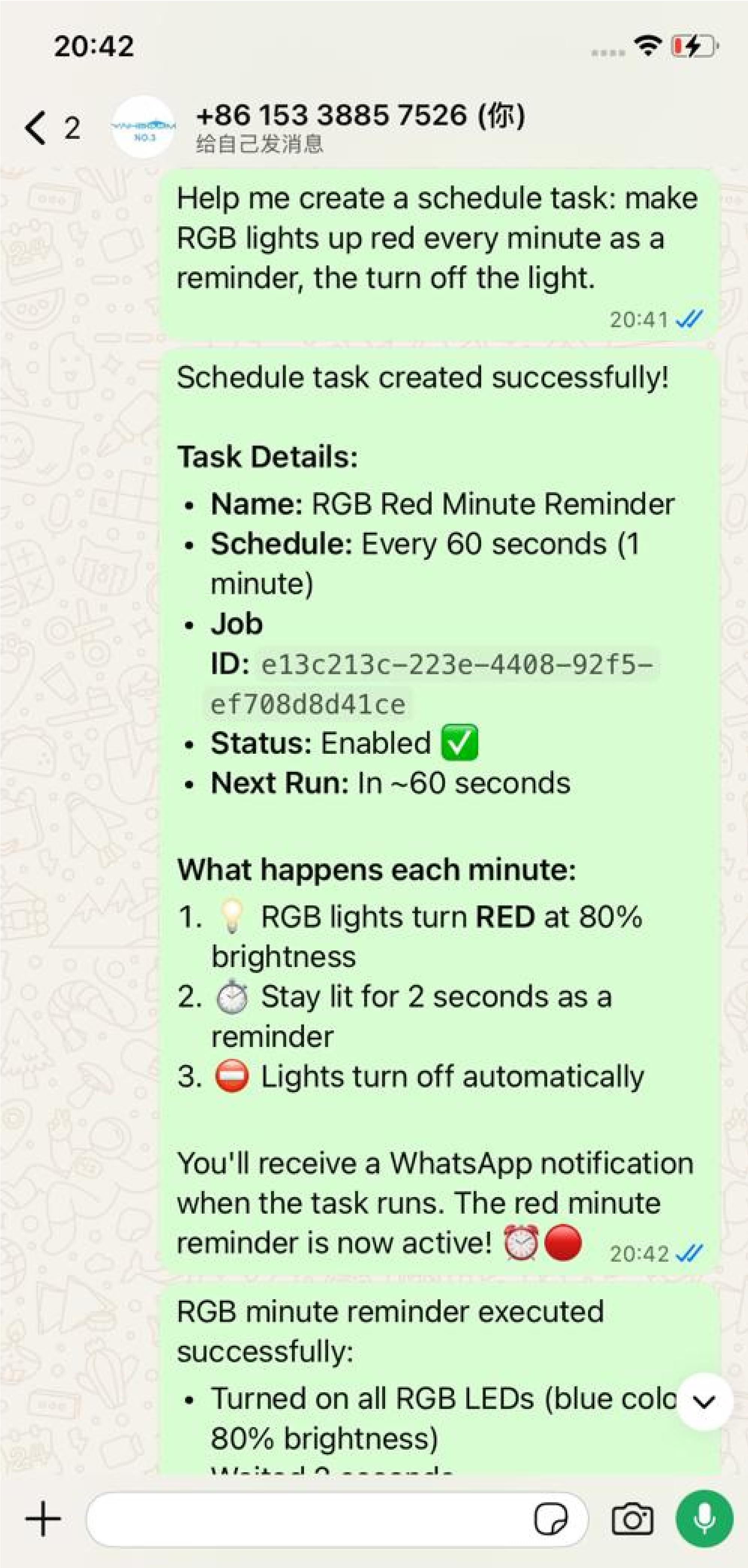
```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) sunrise@ubuntu:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/sunrise/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

4.1 WhatsApp solution

For WhatsApp access steps, please refer directly to [06-Configuration Test of Three Dialogue Schemes]. This section is more suitable for describing requirements, such as:

- Help me do a scheduled task: make the RGB light red every minute as a reminder and then turn off the light


20:42
2
+86 153 3885 7526 (你)
给自己发消息
Help me create a schedule task: make RGB lights up red every minute as a reminder, the turn off the light.
20:41
Schedule task created successfully!
Task Details:
• Name: RGB Red Minute Reminder
• Schedule: Every 60 seconds (1 minute)
• Job ID: e13c213c-223e-4408-92f5-ef708d8d41ce
• Status: Enabled ✅
• Next Run: In ~60 seconds
What happens each minute:
1. 💡 RGB lights turn RED at 80% brightness
2. ⏱️ Stay lit for 2 seconds as a reminder
3. ⛔ Lights turn off automatically
You'll receive a WhatsApp notification when the task runs. The red minute reminder is now active! ⏰🔴
20:42
RGB minute reminder executed successfully:
• Turned on all RGB LEDs (blue colo
80% brightness)

4.2 TUI scheme

If you have completed the TUI self-test in Part 6, enter the terminal text dialogue directly:

```
openclaw tui
```

Then you can directly enter project-related instructions, such as:

- Help me do a scheduled task: read the temperature and humidity every minute and display it on the OLED

```
Help me create a scheduled task: read temperature and humidity every minute
and display on OLED

Scheduled task created successfully!

Task Details:
- Name: Temperature Humidity Dashboard
- Schedule: Every 1 minute (60,000ms)
- Job ID: a43df4b4-6c47-464e-8cce-95ed29bc8a36
- Status: Enabled ✅
- Next Run: In ~60 seconds

What it does:
- Reads temperature and humidity from the DHT11 sensor
- Displays the values on the OLED screen
- Runs automatically every minute in the main session

To manage this task:
- View all jobs: Use cron list command
- Remove job: Use cron remove with the job ID above
- Pause job: Use cron update to set enabled=false

The temperature dashboard will now update automatically every minute!
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
```

4.3 Voice code scheme

```
Requires optional ai voice interaction module + speaker
```

For voice module installation, environment configuration and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration]. After waking up, you can directly say the control content, such as:

- Help me do a scheduled task: read the temperature and humidity every minute and display it on the OLED

```
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Help me create a scheduled task, read temperature and humidity every minute
    and display on aled.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] ## ✅ Scheduled Task Created!

**Task Details:**
- **Name:**
          Temperature Humidity Monitor
- **Job ID:**
          `1a6954fc-f6e9-4aed-8e2f-a5737df6072d`
- **Schedule:** Every 60
          seconds (1 minute)
- **Status:** Enabled ✅
- **Next Run:** In ~60
          seconds

**What It Does:**
1. Reads temperature and humidity from
          the DHT11 sensor
2. Displays the values on the OLED screen
3.
          Repeats automatically every minute

**To Manage This Task:**
-
```

- If you want to close the scheduled task, you can ask Help me stop the current scheduled task in the current conversation.

Stop the current scheduled task for me

The scheduled task "RGB red light every-minute reminder" has been stopped and deleted.

- You can use the command cron list to view the current scheduled tasks

cron list

There are currently no scheduled tasks.

The list is empty, and the previous "RGB red every-minute reminder" task has been succe ly deleted.

5. Project implementation

Note: The following are all dialogue demonstrations in `openclaw tui` mode. You can also choose WhatsApp or voice mode to complete it.

5.2 Scenario 1: Hourly temperature and humidity sampling

Experimental goals

Let the device automatically read temperature and humidity every hour and record the results or sync to the screen.

Effect description

This is the most basic and common scheduled task scenario, and can be expanded into logs, dashboards, environmental alarms, or plant care projects later.

Example instructions

- Help me do a scheduled task: read the temperature and humidity every minute and display it on the OLED

5.3 Scenario 2: Scheduled status reminder

Experimental goals

Use RGB or OLED to make a clear reminder at a fixed time.

Effect description

This type of task is suitable for drinking water reminders, class reminders, inspection reminders, and status switching reminders. It does not emphasize continuous operation, but emphasizes triggering a clear action at a certain point in time.

Example instructions

- Help me do a scheduled task: make the RGB light red every minute as a reminder and then turn off the light

5.4 Scenario 3: Regular inspection script

Experimental goals

String together multiple peripheral actions to form a small automation task that can be executed periodically.

Effect description

For example, the temperature and humidity are read once every hour, and if the status is abnormal, a light will be lit to remind, and the results will be displayed on the OLED. This type of task is close to a complete small project, and it is also the most obvious place where "AI generation script" and "timed execution" are combined.

Example instructions

- Help me do a scheduled task: read the temperature and humidity once every minute, light up the red light when abnormal, and prompt on the OLED